



What is Formula Student?

Formula Student is an **international competition** that brings together the **best engineering schools** and challenges teams made up exclusively of students to **conceive, design, manufacture, and compete** with an electric single-seater built **from scratch**.

In Spain, Formula Student takes place at the iconic **Circuit de Montmeló** in Barcelona and is held every August.

The competition is divided into **two** distinct **parts**: **static events** and **dynamic events**. Between these, teams must **pass a technical inspection** with over a thousand compliance items.

The **static events** focus on engineering **design** justification, **manufacturing cost** analysis, and the development of a **business plan**.

The **dynamic events** are divided into five tests: **acceleration**, **skidpad**, **autocross**, **endurance**, and **efficiency**. These tests aim to push the single-seater to its limits under different scenarios, exploring its top speed, cornering stability, efficiency, and range.



Our History



Our evolution

IFS-01

In just one year, the first team members successfully developed the first prototype in our history: The IFS-01.

With it, they participated in our first competition, Formula Student Spain, in Montmeló. **This single-seater laid the foundation upon which the project would evolve in the following years.**

Competing with the IFS-01 also provided **significant earning and experience** for the team, as they came into **contact with the world of Formula Student for the first time** and had the opportunity to learn from the cars presented by other teams.



IFS-02

With the first single-seater, the team welcomed more members, which **required an organizational shift**. ISC created **management departments** to better support the Formula Student Team and improve its operations. With a larger and better-organized team, the IFS-02 was developed, showcasing notable technical advancements but still leaving room for improvement.

A new **battery architecture** was introduced, similar to the one used by Tesla in their models, based on cylindrical cells.



Our evolution

IFS-03

In the midst of the project, a pandemic disrupted normal operations. However, the team, now composed of over **60 members**, adapted to alternative working methods, allowing the project to move forward with the goal of achieving the first fully functional car.

August 2021 marked a milestone in the team's history with the completion of a fully functional car for the first time. In competition, an outstanding result was achieved in the **Business Plan** event, securing **second place**. Additionally, the excellent feedback received from the judges regarding both mechanical and electrical design provided the team with absolute confidence and enthusiasm heading into 2022.

IFS-04

Not without challenges, the season was envisioned as an **evolution of the previous prototype**, el IFS03. This strategy allowed the team to focus on development in a way that **paid attention to every detail**, ensuring strong performances in competitions such as Formula ATA Italy 2022 and FS Spain 2022.

By the end of the season, the hard work and effort of the entire year paid off. **For the first time, the team successfully passed scrutineering** (Technical inspections) in both competitions attended during the summer and **participated in Dynamic events at both**.



Our evolution

IFS-05

Despite significant challenges at the beginning of the season—including the largest generational turnover the project had ever faced and numerous structural changes to the single-seater, the ISC achieved its best-ever competition results.

After competing in Formula ATA Italy 2023, doubling our points in static events, securing P2 in the business Plan, and climbing three positions compared to 2022, we attended Formula Student Spain. There, we participated in all the dynamic events for the first time in our history, achieving results that marked a turning point for the team: P6 in Business Plan, P6 in Cost Event, P9 in Design Event, DNF in Skidpad, P6 in Acceleration, P9 in Autocross, P6 in Endurance y P5 in Efficiency, achieving P7 in the Overall Ranking.

IFS-06

The 2023/2024 season leaves us with a sense of accomplishment, marking the **third time we participated in dynamic events** and solidifying our international presence.

We faced significant challenges in the early competitions of the season, such as Formula Student Spain. Additionally, we competed for the first time in **Formula Student Germany**, the flagship competition in the electric category.

The season concluded at Formula Student Italy, where we participated in the **dynamic events and achieved the first-ever podium** in the Business Plan. The IFS-06 overcame harsh weather conditions, **running an endurance race in the rain for the first time**, a mayor achievement for the team given the risks involved.



Our evolution

IFS-07: A new beginning

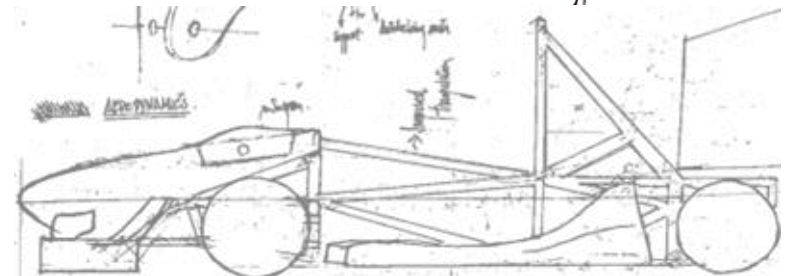
The 2024/2025 season leaves us with the need **to keep pushing**, marking the **fourth time we pass all the technical inspections**.

This year marks a new beginning, bringing again a new prototype, this time with an **aerodynamic package** and an overall optimization of components, which led to the **fastest, and lightest car of our history**.

Due to technical difficulties in the endurance, we were not able to finish the competition. However, the team delivered during the static events and throughout the whole competition.

This events made us **stronger** to build a better car for the next season, aiming to **compete in the best competitions** around Europe.

07 - An advisory to the impossible



Our evolution

IFS-08: A milestone

The 2025/2026 season confirms the maturity of the project. ISC competes in two major European events — Formula Student Austria, at the **Red Bull Ring** in Spielberg, and Formula Student Spain, at the **Circuit de Barcelona-Catalunya**.

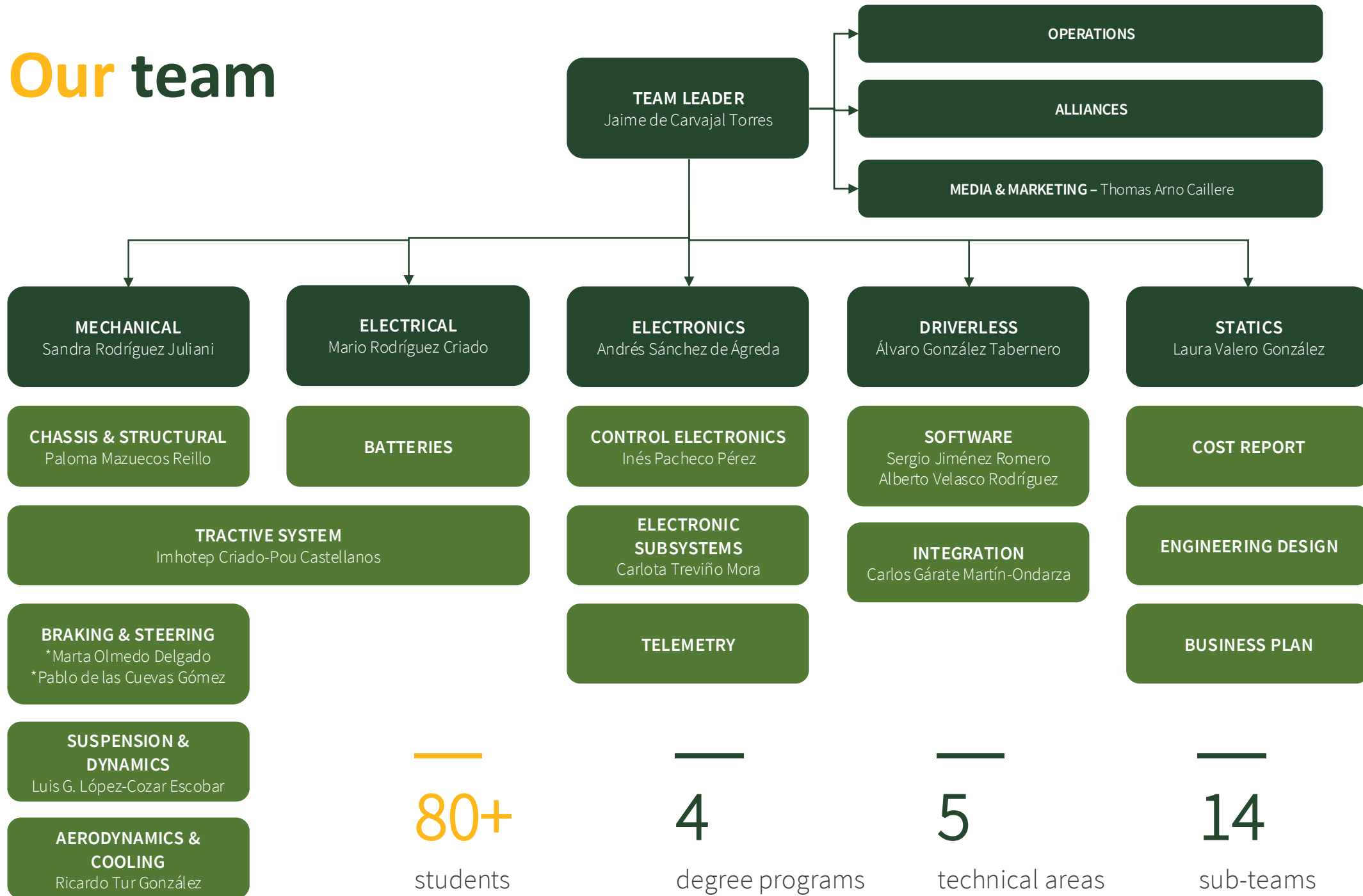
In both competitions, the team completes all dynamic events for the first time in its history: skidpad, acceleration, autocross, and endurance, demonstrating the reliability of the IFS-08 across two very different tracks and conditions.

This season also marks a technological milestone: the integration of the first **driverless** developments within the project, laying the groundwork for the autonomous system the team will continue developing in upcoming seasons.

The IFS-08 thus establishes itself as the most reliable car in ISC's history.



Our team



Season 2026/2027

Promising outlook

Having completed all dynamic events in Austria and Spain with the IFS-08 allows the team to approach the 2026/2027 season with a renewed vision, focused on two major leaps: traction and autonomy.



Generational transition

- New member selection
- Mentoring for knowledge transfer



Vehicle concept

- Two in-wheel motors
- Dual rear-wheel drive
- Stronger structure



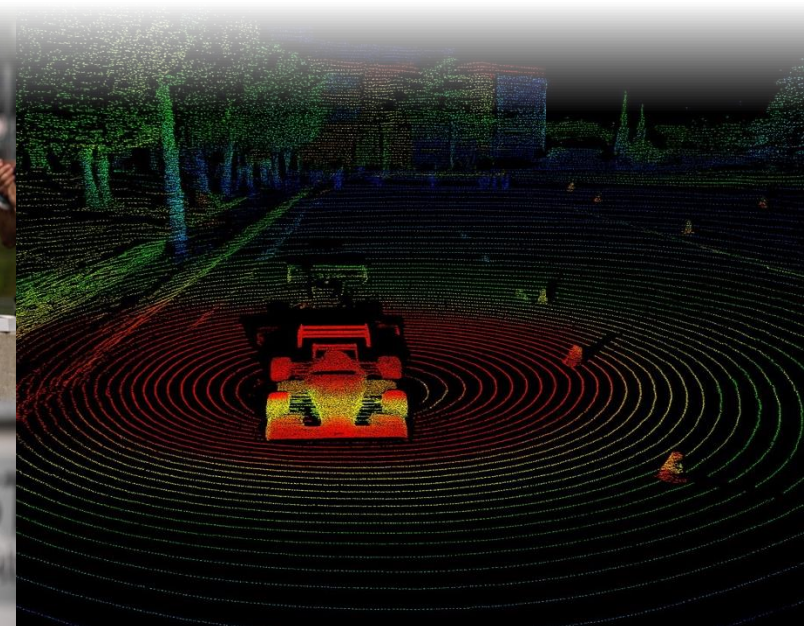
Technical objectives

- Reliability
- Simplification



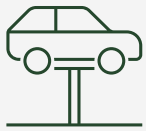
Autonomous functionality

- Debut in dynamic driverless events



Driverless: from integrated to proven

Where the autonomous system stands, and the goal for 2026/2027



LAST SEASON

Driverless running on the car

The autonomous stack was integrated into IFS-08 and validated in testing.

- Perception and SLAM
- Path planning
- Adaptive control



THIS SEASON

A full dynamic driverless event

Finish the test campaign and race the car end to end.

- Complete driverless testing
- Acceleration and skidpad
- Autocross and trackdrive

AS at ISC: our stack



Perception

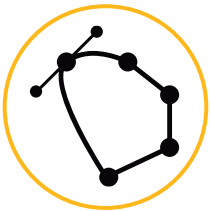
New filtering and terrain understanding of the track

75% fewer false positives



SLAM

Simultaneous localization and mapping straight from the sensors



Path Planning

FaSTTUBe algorithm: optimized trajectories through graph search and interpolation



Control

PID and Stanley adaptive control: steadier speed, direction and lap times

What do we need in order to grow?

1

Team

80+ students giving their free time, passion and effort

2

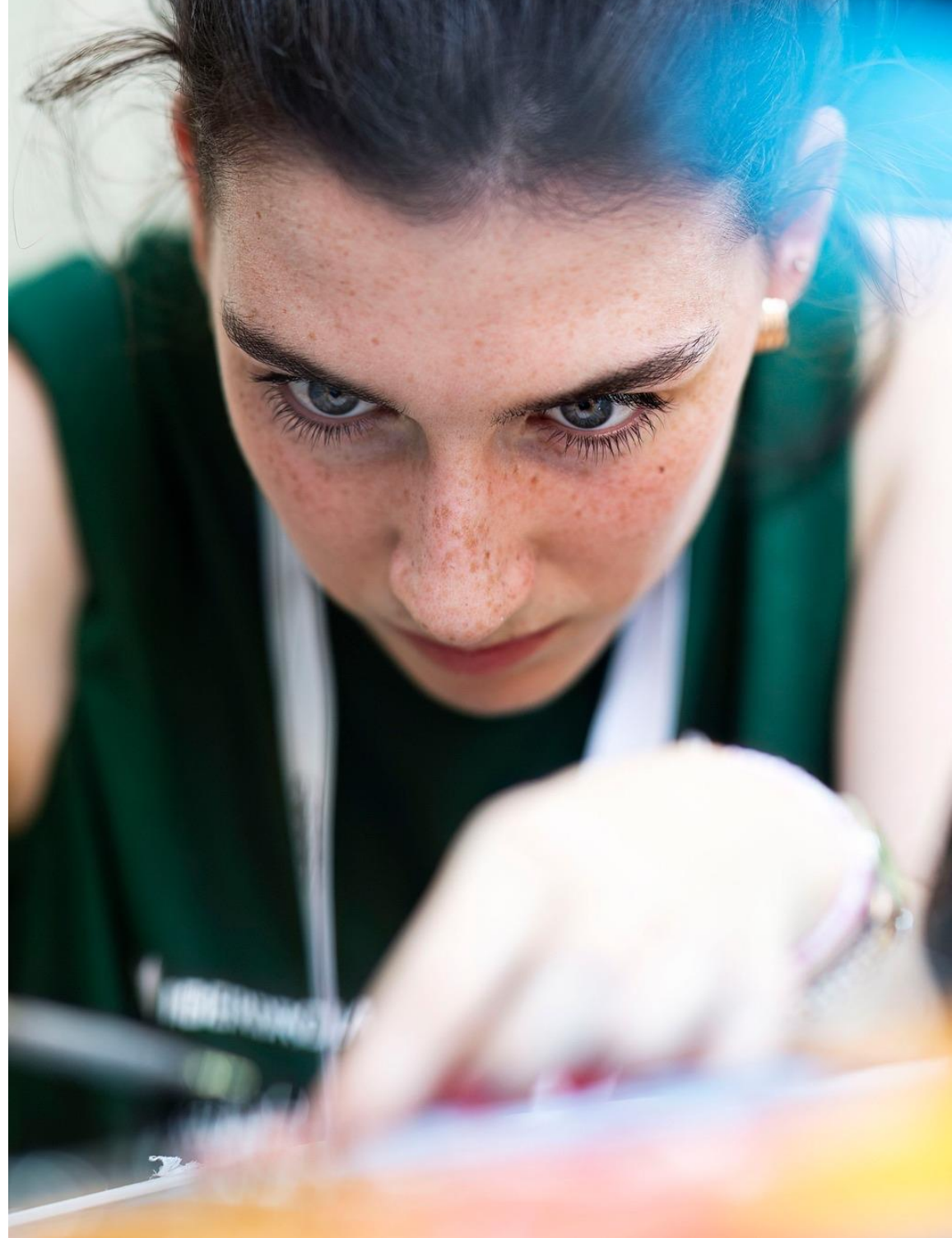
Experience

Seven seasons, from second-year undergraduates to master's students

3

Means

Sponsors turn that team and experience into a faster car



How to become a partner in this project?

We look for companies that share our drive for innovation and see Formula Student as a showroom for young engineering talent.



LOGISTICS PARTNER

Transport, tools and workshop material



FABRICATION PARTNER

Parts and components, or help manufacturing them



FINANCE PARTNER

Funding for specific lines of research inside the project

We look for long-term deals that both sides build on.





What would you be supporting by joining the project?



Young talent

Reports, deadlines and real responsibility before graduating



Innovation in electric mobility

Hands-on training in electric systems



Innovation in autonomous mobility

First autonomous electric efforts in Formula Student Madrid



Coaching Spanish engineers

Backing Spanish universities internationally



NX Technologies



Our
sponsors



PRINCIPIA



ISC Formula Student Racing Team
Season 2026/2027

THANK YOU

Innovation **TOGETHER**

